

HEAT INDEX
(APPARENT TEMP.)

$$I = f(T, H)$$

MULTIVARIABLE FUNCTION OF 2 VARS.

Partial Differentiation

COMBINING EFFECT OF T AND HUMIDITY (H)

Trim Ch 11, S. 12.3

CALCULATING RATE OF CHANGE OF I WHEN $H = 70\%$

→ RIGHT : $\frac{43-41}{1} = 2^\circ\text{C}$

→ LEFT : $\frac{41-39}{1} = 2^\circ\text{C}$

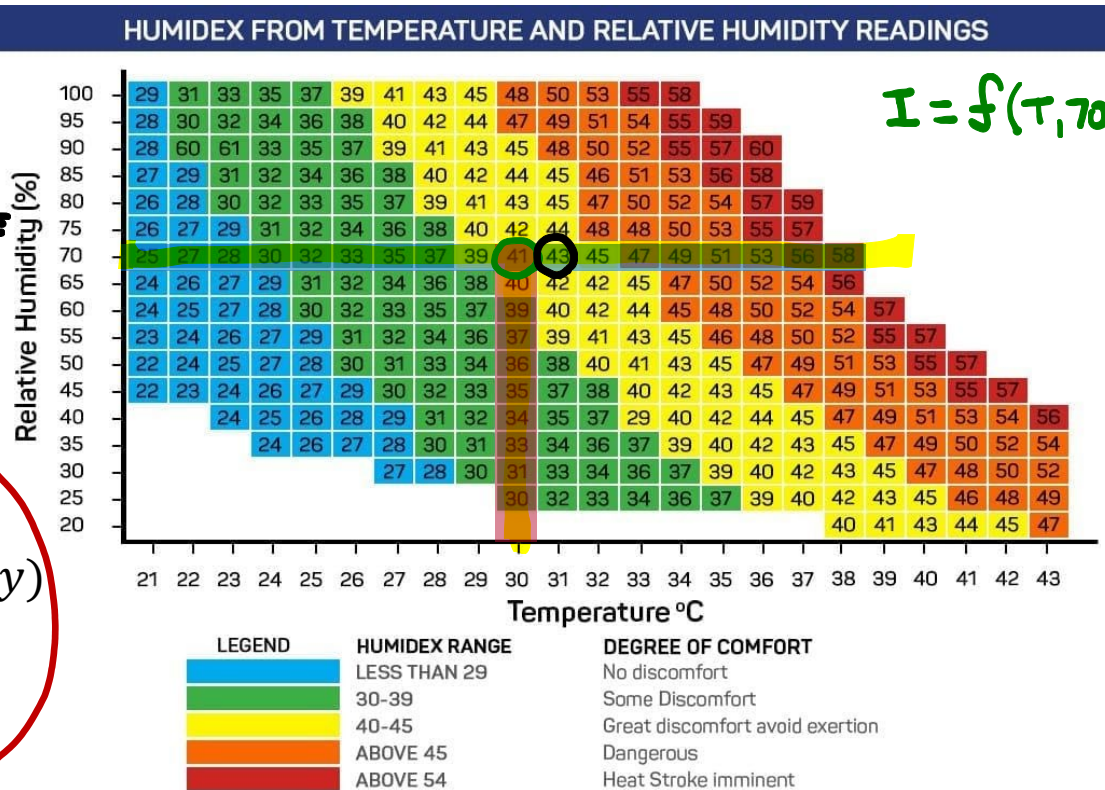
Outline:

1. Partial differentiation of $f(x, y)$
2. Notation (∂)



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"I" RISES BY 2°C
FOR EVERY DEGREE THAT "T" RISES



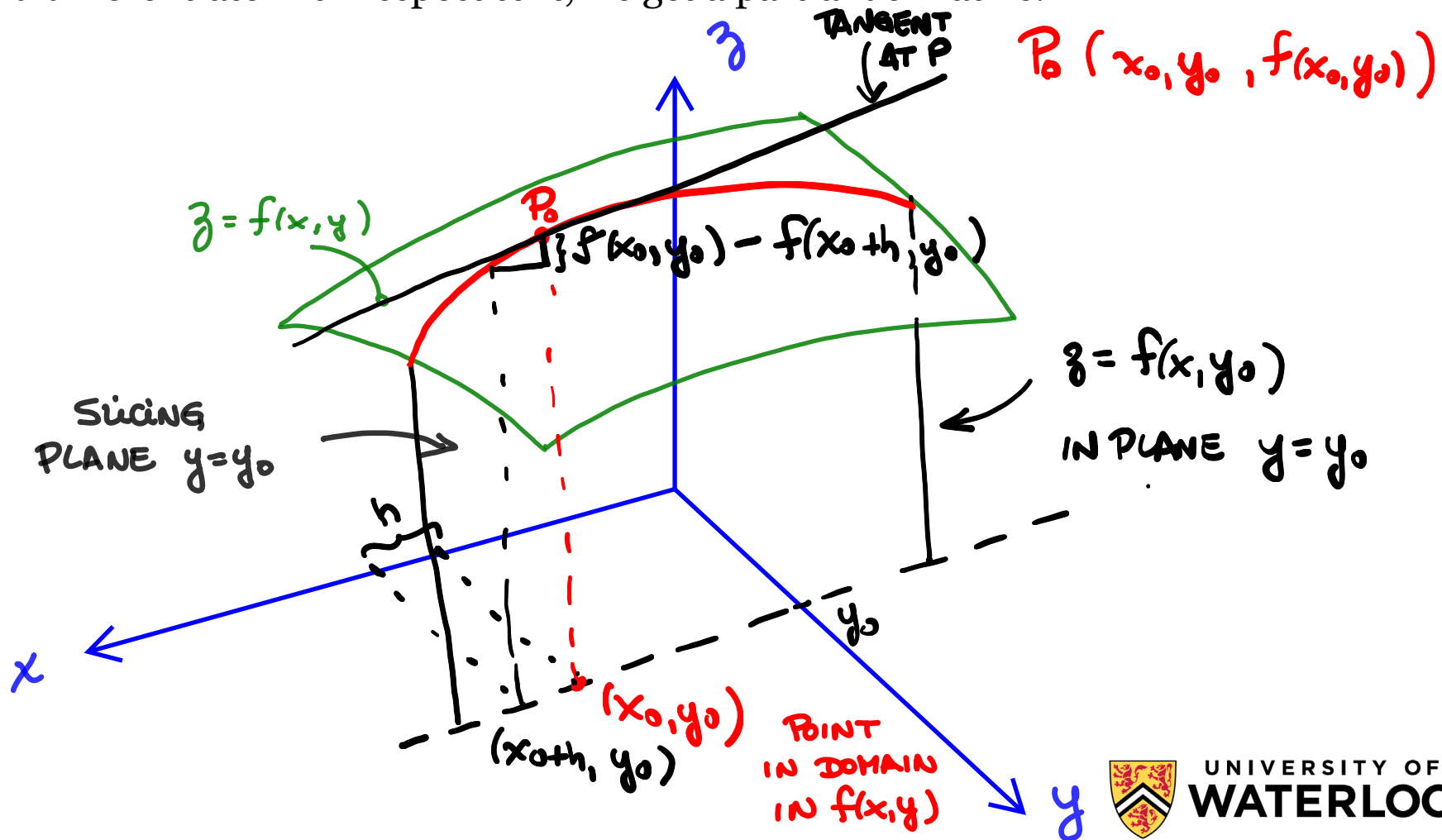
Source: Environment Canada

<https://www.canada.ca/en/environment-climate-change/services/seasonal-weather-hazards/spring-summer.html#humidex>

Source: https://www.ccohs.ca/oshanswers/phys_agents/humidex.html

Partial Derivative of a Function of Two Variables

When we hold all but one of the independent variables of a function constant and differentiate with respect to it, we get a partial derivative.



Partial Derivative of a Function of Two Variables

The slope of the tangent line at P is the rate of change of $f(x, y)$ as we vary x , holding y constant.

This is known as the first order partial derivative and provided that the limit exists, it can be calculated as,

$$\left. \frac{\partial f}{\partial x} \right|_{(x_0, y_0)} = \lim_{h \rightarrow 0} \frac{f(x_0 + h, y_0) - f(x_0, y_0)}{h}$$

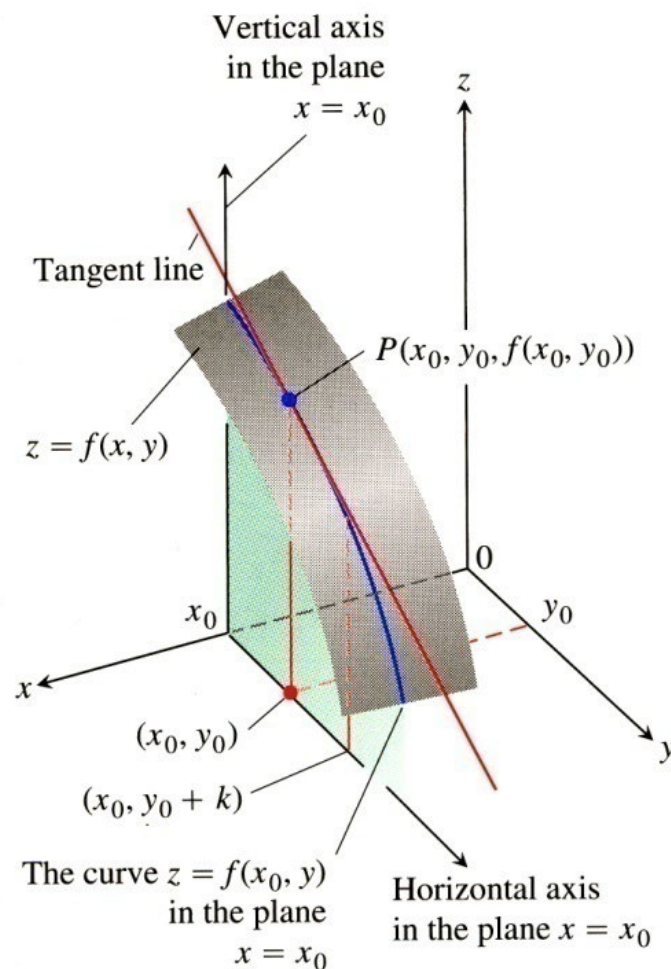
The symbol $\frac{\partial}{\partial x}$ is read as “partial derivative” with respect to x .

Partial Derivative of a Function of Two Variables

We can write an analogous expression for the partial derivative of $f(x, y)$ with respect to y .

The first Order Partial Derivative of a function $f(x, y)$ with respect to y holding x constant is then,

$$\left. \frac{\partial f}{\partial y} \right|_{(x_0, y_0)} = \lim_{k \rightarrow 0} \frac{f(x_0, y_0 + k) - f(x_0, y_0)}{k}$$



Other Notations for First Order Partial Derivatives

First order Partial Derivative of a function $f(x, y)$ with respect to x :

$$\left. \frac{\partial f}{\partial x} \right|_{(x_0, y_0)}, \quad f_x, \quad \frac{\partial z}{\partial x}, \quad z_x, \quad \left. \frac{\partial z}{\partial x} \right|_y, \quad \left. \frac{d}{dx} f(x, y_0) \right|_{x=x_0}$$

First order Partial Derivative of a function $f(x, y)$ with respect to y :

$$\left. \frac{\partial f}{\partial y} \right|_{(x_0, y_0)}, \quad f_y, \quad \frac{\partial z}{\partial y}, \quad z_y, \quad \left. \frac{\partial z}{\partial y} \right|_x, \quad \left. \frac{d}{dy} f(x_0, y) \right|_{y=y_0}$$



Example

If $f(x, y) = \sin\left(\frac{x}{1+y}\right)$, calculate $\frac{\partial f}{\partial y}$ and $\frac{\partial f}{\partial x}$

Solution:

USING RULE OF DIFF.

$$\frac{\partial f}{\partial x} = \cos\left(\frac{x}{1+y}\right) \cdot \frac{\partial}{\partial x}\left(\frac{x}{1+y}\right) = \cos\left(\frac{x}{1+y}\right) \cdot \left(\frac{1}{1+y}\right)$$

$$\frac{\partial f}{\partial y} = \cos\left(\frac{x}{1+y}\right) \cdot \frac{\partial}{\partial y}\left(\frac{x}{1+y}\right) = -\cos\left(\frac{x}{1+y}\right) \cdot \frac{x}{(1+y)^2}$$



Higher Order Partial Derivatives

If $z = f(x, y)$ is a function of 2 variables, then its partial derivatives $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ are also functions of 2 variables, and we can calculate their partial derivatives as,

$$\frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right) = \frac{\partial^2 f}{\partial x^2} \rightarrow (f_x)_x = f_{xx}$$

$$\underbrace{\frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right)}_{\text{LEIBNITZ}} = \frac{\partial^2 f}{\partial y \underbrace{\partial x}_{\text{PERFORMED FIRST}}} \rightarrow (f_x)_y = f_{xy}$$

PERFORMED FIRST

LEIBNITZ



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Higher Order Partial Derivatives

Similarly,

$$\frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) = \frac{\partial^2 f}{\partial x \partial y} \rightarrow (f_y)_x = f_{yx}$$

PERFORMED FIRST

$$\frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) = \frac{\partial^2 f}{\partial y \partial x} \rightarrow (f_x)_y = f_{xy}$$

PERFORMED FIRST

These partial derivatives are called the **second order partial derivatives** of $f(x, y)$.



Higher Order Partial Derivatives

Example:

Calculate the second order partial derivatives of

$$f(x, y) = x \cos y + ye^x$$

FIRST ORDER
PARTIAL DER.

$$f_x = \frac{\partial f}{\partial x} = \cos y + ye^x$$

$$f_y = \frac{\partial f}{\partial y} = -x \sin y + e^x$$

$$\underbrace{\frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right)}_{f_{xx}} = ye^x$$

$$\underbrace{\frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right)}_{f_{xy}} = \underbrace{-\sin y + e^x}_{\text{red bracket}}$$

$$\underbrace{\frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right)}_{f_{yx}} = \underbrace{-\sin y + e^x}_{\text{red bracket}}$$

$$\underbrace{\frac{\partial}{\partial y} \left(\frac{\partial f}{\partial y} \right)}_{f_{yy}} = -x \cos y$$

MIXED DERIVATIVES
ARE EQUAL IN VALUE!



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Mixed Derivative (or Schwarz's) Theorem

If the second partial derivatives of $f(x, y)$ are continuous at (a, b) ,

$$\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial^2 f}{\partial x \partial y}$$

Or, using our other notation,

$$f_{xy} = f_{yx}$$

These derivatives are referred to as mixed derivatives. While they may be denoted differently, they are equal in value.

Higher Order Partial Derivatives

Partial derivatives of order 3 can also be defined,

$$\frac{\partial}{\partial x} \left(\frac{\partial^2 f}{\partial^2 y} \right) = \frac{\partial^3 f}{\partial x \partial^2 y} = f_{yyx}$$

The mixed derivative theorem also applies for partial derivatives of order 3. For example,

$$f_{yyx} = f_{yx y} = f_{xyy}$$

And for functions of more than 3 variables,

$$f_{xx y z r y y} = f_{xy x z y y r}$$

The only requirement is to differentiate with respect to each variable the same number of times!



Summary

- We learned the concept of partial differentiation
- We learned that to calculate partial derivatives we use the same set of rules for finding the derivative of a function of one variable (good news!)
- When we differentiate a function $f(x, y)$ twice, we produce second order partial derivatives.
- We have learned a second notation (other than the Leibnitz notation) to denote higher order partial derivatives.
- The mixed derivative (or Schwartz's) Theorem states that $f_{xy} = f_{yx}$ for continuous partial derivatives, and it also works for higher order partial derivatives
 - The only requirement is to differentiate with respect to each variable the same number of times!

